

ASHIK KANNAMPILLY JANARDHANAN

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PROFESSIONAL SUMMARY

MSc Artificial Intelligence and Machine Learning graduate (First Class Honours, 1.1, University of Limerick) and Computer Science graduate with two years of industry experience at Cyber Prism Software Solutions. Develops and validates machine learning algorithms in Python (Scikit-learn, PyTorch, TensorFlow) for supervised classification with neural networks and tree-based models, including feature engineering, performance validation and statistical significance testing. Also writes C and C++ and writes up results clearly.

EDUCATION

MSc in Artificial Intelligence and Machine Learning: First Class Honours (1.1) Sep 2025 - Sep 2026
University of Limerick, Ireland

Core modules: Advanced AI Ecosystems, Deep Reinforcement Learning, Cloud Computing, Machine Learning

Final year project: RETINEXPLAIN: Explainable Retinal Disease Detection (MSc dissertation)

B.Tech in Computer Science (Artificial Intelligence): CGPA 9.08/10 Dec 2020 - Jun 2024
APJ Abdul Kalam Technological University, Kerala, India

Core modules: Algorithms and Data Structures, Object-Oriented Programming, Software Engineering, Distributed Systems, Operating Systems

WORK EXPERIENCE

Cyber Prism Software Solutions | Muvattupuzha, Kerala, India Jun 2023 - Jun 2025
AI/ML Engineer Jul 2024 - Jun 2025

- Machine learning:** Developed, trained, evaluated and optimised machine learning and deep learning models in PyTorch and Scikit-learn.
- Data pipelines:** Designed ETL pipelines in Python and SQL to extract, clean and load data into Snowflake for data warehousing and analysis.
- LLM applications:** Built LLM-powered features with the OpenAI API and integrated the model outputs into Python applications.
- Cloud and version control:** Deployed and ran ML solutions on AWS, with code managed through Git and GitHub in a shared team workflow.
- Reporting:** Built Power BI and Tableau dashboards that presented model results and key metrics to the team.

AI/ML Intern Jun 2023 - Jun 2024

- Data analysis:** Cleaned, explored and analysed datasets with Pandas, NumPy and Excel to prepare data for model training.
- SQL and visualisation:** Wrote SQL queries to pull and transform data, and built Power BI and Tableau visualisations to share findings.
- Machine learning:** Built and evaluated models with Scikit-learn on project work and technical assignments.

Intern: Essentials of Ethical Hacking | Edwhere Education Internship Program May 2023

- Scored 96% in a practical assessment on IAM (identity and access management) concepts, network scanning and steganography, using Nmap and Steghide for penetration testing and vulnerability assessment on TryHackMe.

ML Lead Intern | Suyati Technologies Ltd, Kochi, India Mar 2022 - Sep 2022

- Collaborated closely with the team to match model outputs to product requirements and operational workflows, as the ML owner in a 3-person team (ML, front-end, back-end) building an airline booking system.
- Built an airline delay prediction model with XGBoost on structured aviation datasets, then trained and validated it and delivered it as a pickle file to the back-end teammate for integration into the live application, enabling real-time inference.

Data Science Intern | Capabl (NASSCOM Recognised) Jan 2022 - May 2022

- Completed a 22-week accredited curriculum in Python, SQL, ML algorithms and deep learning, and built and evaluated classification models with structured metric reporting.

KEY PROJECTS

RETINEXPLAIN: Explainable Retinal Disease Detection | *MSc dissertation, explainable AI* 2026

- Improved accuracy from 46.16% to 56.28% over the strongest simple interpretable baseline at a comparable macro F1 of 0.4243 (paired significance test across 30 runs, $p < 0.0001$; ODIR-5K, 5,684 images), with decisions that can be audited through symbolic rules instead of a black box.

Crop Disease Detection: Serverless ML Pipeline on AWS | *Python, SageMaker, 14 AWS services* 2026

- Raised classification accuracy from 81% to 94.17% across 38 classes on 3,582 test images, with zero errors across all production invocations, by engineering 51 visual and 15 NLP features.
- Designed and deployed the pipeline across 14 AWS services (API Gateway, Lambda, Rekognition, Comprehend, SageMaker-hosted XGBoost, DynamoDB, S3, SNS, CloudWatch), with an Application Load Balancer across 3 availability zones and an Auto Scaling Group for high availability.

Financial Loan Risk Analytics | *Python, Pandas, Scikit-learn, Power BI* **2026**

- Reached 92.53% accuracy and an 80.84% F1 score on loan outcome prediction, by cleaning and analysing 32,000+ loan records and building a Random Forest classifier, with the predictions shown in an interactive Power BI dashboard.

Global E-commerce Sales Analytics | *Snowflake, SQL, Tableau* **2026**

- Built a cloud-based analytics pipeline in Snowflake and found sales patterns across countries, product categories and months in about 1,200 order records, with SQL views and an interactive Tableau dashboard.

PolicyPilot: RAG-Based Company Policy Assistant | *Python, LangGraph, OpenAI API, ChromaDB, Streamlit* **2026**

- Built a single-agent assistant that answers company policy questions with retrieval augmented generation, using PyMuPDF, text chunking and embeddings stored in the ChromaDB vector database.
- Added tool calling for calculations and policy comparisons, source citations, handling of unsupported queries, and a Streamlit interface for document upload.

Financial Risk Assessment Agent | *Python, LangGraph, LangChain, Groq API* **2026**

- Built an agent in Google Colab on the Groq API (Llama 3.3 70B), orchestrated with LangGraph and LangChain tool calling, designed to reason over financial risk queries step by step instead of returning a single fixed response.

Other projects: DQN agent for Atari Boxing (TensorFlow, Keras, Gymnasium); Animal Detection and Alert System (YOLOv5, Gmail API).

TECHNICAL SKILLS

AI and LLMs: LLMs and the OpenAI API, RAG, AI agents (LangGraph, LangChain tool calling), embeddings and vector search (ChromaDB), Scikit-learn, XGBoost, PyTorch, TensorFlow, Keras

Programming: Python, C++, C, Java, SQL, R, JavaScript (familiar)

Data: Snowflake, MySQL, ETL, data warehousing, Pandas, NumPy, Power BI, Tableau

Cloud and APIs: AWS (Lambda, API Gateway, SageMaker, EC2, S3, DynamoDB, SNS, IAM, CloudWatch); Azure ML (basic); REST APIs (API Gateway, FastAPI familiar)

Engineering practice: Git, GitHub, CI/CD concepts, Agile/Scrum, access control (IAM least privilege design), Jupyter/Google Colab, Streamlit

Currently learning: digital signal processing basics (filtering and FFT with SciPy); MATLAB basics; classifying accelerometer data from a public sensor dataset; linear algebra and probability refresh; embedded C basics

PUBLICATION AND CERTIFICATIONS

Publication: Next-Generation Cross-Language Visual Transit Navigation System Leveraging Deep Learning and NLP, Springer International Conference Proceedings, Singapore, March 2025.

Certifications: Data Analytics Fundamentals (AWS, 2023); Introduction to AI (IBM, 2025); Data Analytics (GrowAI, 2025); Google Data Analytics (Coursera, 2023); Python (Coursera, 2023).